



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2002
TRIAL HIGHER SCHOOL
CERTIFICATE EXAMINATION

Mathematics

General Instructions

- Reading time – 5 minutes.
- Working time – 3 hours
- Write using black or blue pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may not be awarded for messy or badly arranged work.

Total Marks - 120 marks

- All questions are of equal value.

Examiners: *P. Bigelow, P. Parker*

NOTE: This is a trial paper only and does not necessarily reflect the content or format of the final Higher School Certificate Examination Paper for this subject.

Question 1: (12 Marks)**Marks**

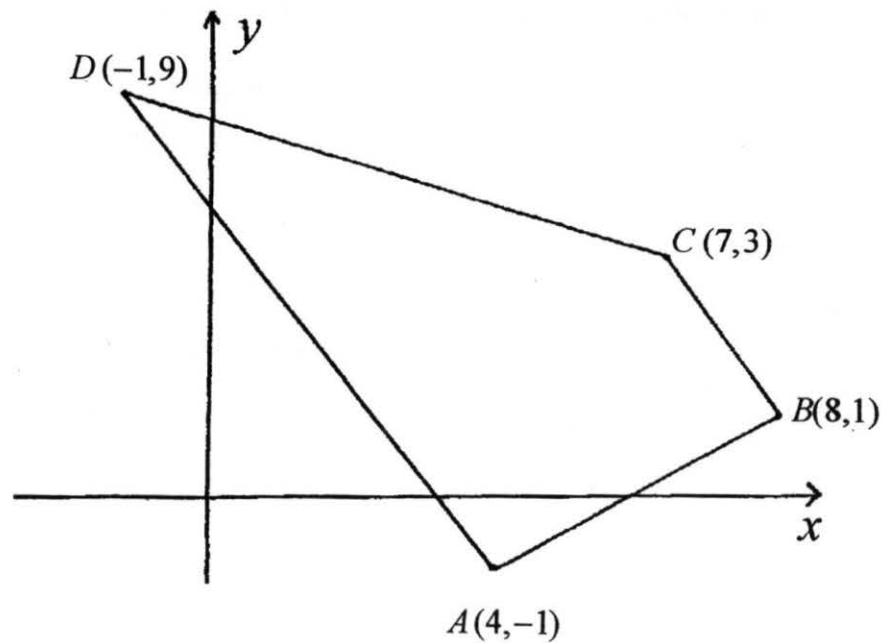
- (a) Solve $\frac{3}{x} = \frac{4}{5}$ 2
- (b) If $a = 2.673$ and $b = 1.049$, evaluate $\frac{a^2 + b^2}{ab}$ correct to 2 decimal places. 2
- (c) Factorise $cd - c - dy + y$ 2
- (d) The line $kx - y = 29$ passes through the point $(4, -1)$, find the value of k . 2
- (e) Graph the solution to $|x + 3| \leq 1$ on a number line. 2
- (f) Find integers a and b such that 2

$$\frac{1}{\sqrt{3} + 2} = a\sqrt{3} + b$$

START A NEW BOOKLET

Question 2: (12 Marks)

Marks



A , B , C and D are the points $(4, -1)$, $(8, 1)$, $(7, 3)$ and $(-1, 9)$ respectively.

- | | | |
|-----|--|---|
| (a) | Show that the equation of AC is $4x - 3y - 19 = 0$ | 2 |
| (b) | Show $BC \parallel AD$ | 2 |
| (c) | Show $\angle ACD = 90^\circ$ | 2 |
| (d) | Show the length of AC is 5 units | 2 |
| (e) | Find the perpendicular distance of B from AC | 2 |
| (f) | Find the area of the trapezium $ABCD$. | 2 |

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Question 3: (12 Marks)

Marks

(a) Differentiate

(i) $(2x-1)^7$ 1

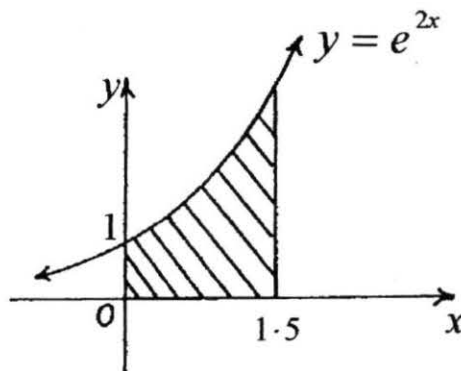
(ii) $\log_e(4+5x)$ 2

(iii) $x^2 \sin x$ 2

(b) Find $\int \sec^2 3x \, dx$ 1

(c) Evaluate $\int_1^{e^2} \frac{3}{x} \, dx$ 3

(d) 3



The diagram above shows the region bounded by the curve $y = e^{2x}$, the line $x = 1.5$ and the coordinate axes.
Find the EXACT area of this region.

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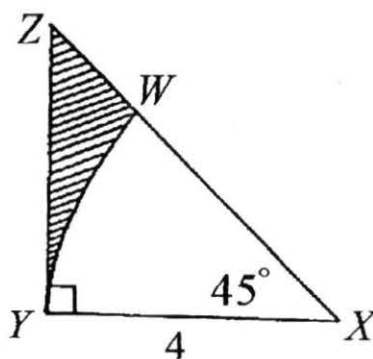
Question 4: (12 Marks)

Marks

- (a) A parabola has equation $x^2 - 2x + 25 = 8y$.
- (i) By completing the square, express this in the general form $(x - p)^2 = 4a(y - q)$ 2
- (ii) State the coordinates of the vertex. 1
- (iii) State the coordinates of the focus. 1
- (iv) State the equation of the directrix. 1
- (b) Find the equation of the line through the intersection of the lines $x - y = 0$ and $x + 2y - 6 = 0$ and which is parallel to the line $3x - 2y + 7 = 0$. 4

Leave your answer in general form.

- (c) 3



In the diagram above, XYZ is a right-angled triangle, $\angle ZXY = 45^\circ$ and $XY = 4$ units. A circular arc, centre X and radius YX cuts the side XZ at W .

Find the EXACT area of the shaded region WZY .

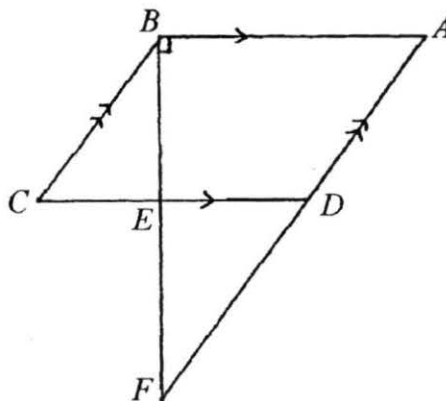
START A NEW BOOKLET

Question 5: (12 Marks)

Marks

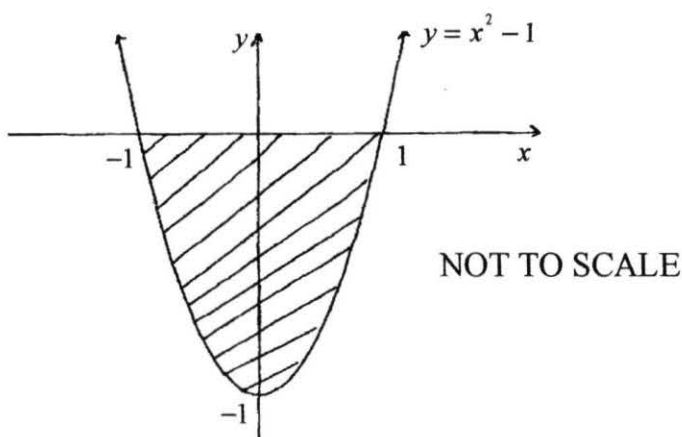
- (a) Find the equation of the normal to $y = x^2 - 3x + 5$ at the point $(3, 5)$.
Leave your answer in general form. 2

(b)



$ABCD$ is a parallelogram and $FB \perp AB$

- (i) Prove $\triangle CBE \parallel \triangle AFB$ 3
- (ii) If $CE = 3$ cm, $BC = 7$ cm and $AF = 15$ cm, find AB . 2
- (c) 3



The area bounded by the curve $y = x^2 - 1$ and the x -axis is rotated about the x -axis.
Find the volume of the solid of rotation. Leave your answer in EXACT form.

- (d) A continuous curve $y = f(x)$ has the following properties for the closed interval $x_1 \leq x \leq x_2$: 2

$$f(x) > 0, \quad f'(x) < 0, \quad f''(x) > 0$$

Sketch a curve satisfying these conditions.

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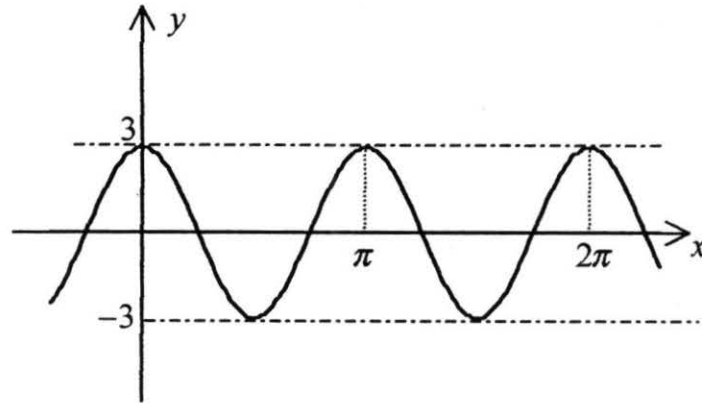
Question 6: (12 Marks)

Marks

- (a) The curve $y = x^3 + mx + n$ has a stationary point at $P(1,5)$. Find the values of the constants m and n .

3

(b)



The graph above represents $y = a \cos mx$.

2

Write down the values of a and m .

- (c) Express as a single logarithm in simplest form

2

$$\ln 2 + 2 \ln 18 - \frac{3}{2} \ln 36$$

- (d) Simplify $\sqrt{\frac{1 - \cos^2 \theta}{1 + \tan^2 \theta}}$

2

- (e) The table shows the value of a function $f(x)$ for five values of x .

3

x	-1	1	3	5	7
$f(x)$	5	9	2	-1	-6

Use Simpson's rule with these five values to estimate $\int_{-1}^7 f(x) dx$

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Question 7: (12 Marks)

Marks

- (a) The value $\$V$ of a particular make of car can be calculated using the equation

$$V = 50000e^{-0.25t}, \text{ where } t \text{ years is the age of the car}$$

- | | | |
|-------|--|---|
| (i) | State the value of the car when it is new | 1 |
| (ii) | What is the value of the car after 3 years? (to the nearest \$100). | 2 |
| (iii) | How long will it take for the car to be worth \$15 000?
(Answer to the nearest tenth of a year) | 2 |

- (b) A total of \$15 600 is to be shared among three people A, B and C. If A is to get the smallest share of \$1600, find the value of the remaining shares when

- | | | |
|------|--|---|
| (i) | the values of the three shares form an arithmetic series | 2 |
| (ii) | the values of the three shares form a geometric series | 2 |

- (c) Two cards are chosen at random, without replacement from the seven cards below.

2	2	4	6	6	6	6
---	---	---	---	---	---	---

What is the probability that

- | | | |
|------|--|---|
| (i) | both cards are 2? | 1 |
| (ii) | the sum of the two numbers on the cards chosen, is greater than 8? | 2 |

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Question 8: (12 Marks)

Marks

- (a) Two artillery guns are situated 3 kilometres apart at positions X and Y respectively. They are both aiming at a target T . 3

The angles XYT and YXT are respectively 72° and 78° .

Find the distance between the target and the gun nearer to it

- (b) Given that $x = -\frac{1}{2}$ is one root of the following quadratic equation 3

$$mx^2 - 20x + m = 0,$$

find the exact value of the other root.

- (c) Consider the function $y = x^3 + 3x^2 - 9x$

- | | | |
|-------|--|---|
| (i) | Find the coordinates of the stationary points. | 2 |
| (ii) | Find the domain in which the curve is concave upwards. | 1 |
| (iii) | Sketch the curve for $-5 \leq x \leq 3$. | 3 |

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Question 9: (12 Marks)

Marks

(a) Solve $|x + 1| = 2x + 7$

3

- (b) Comment on the following reasoning ie JUSTIFY whether the following statement is TRUE or FALSE

2

“When two coins are tossed, they can either fall as two heads or two tails or as a tail and a head.

As there are three possibilities the probability of 2 heads is $\frac{1}{3}$.”

- (c) The acceleration a metres per second per second of a moving object is given at time t seconds ($t \geq 0$) by

$$a = 2\pi^2 \cos \pi t$$

At time $t = 0$, the object is at the point $x = 0$, and travelling with velocity $v = \pi$ metres per second.

- (i) Find the velocity v and the displacement x as a function of t , for $t \geq 0$
- (ii) Find, for t in the range $0 \leq t \leq 4$, the values of t for which the object is stationary.
- (iii) Show that, for t in the range $2 \leq t \leq 4$, the largest value of x is

3

2

2

$$2 + \sqrt{3} + \frac{19\pi}{6}.$$

START A NEW BOOKLET

Question 10: (12 Marks)

Marks

(a) $C(n) = 1000(0.8)^n + 10\,000[1 + 0.8 + (0.8)^2 + \dots + (0.8)^{n-1}]$

3

Find the limiting sum of C as the number of terms increases indefinitely

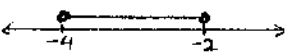
- (b) A scientist has found that the amount, $Q(t)$, of a substance present in a mineral at time $t \geq 0$ satisfies

$$4 \frac{d^2 Q}{dt^2} + 4 \frac{dQ}{dt} + Q = 0$$

- (i) Verify that $Q(t) = A(1+t)e^{-0.5t}$ satisfies this equation for any constant $A > 0$ 4
- (ii) If $Q(0) = 10$ mg, find the maximum value of $Q(t)$ and the time at which this occurs. 4
- (iii) Describe what happens to $Q(t)$ as $t \rightarrow \infty$ 1

THIS IS THE END OF THE PAPER

SBHS MATHEMATICS TRIAL 2002

①		2(a)	
(a)	$4x = 15$ $x = 1\frac{3}{4}, 3\frac{3}{4}$	2	$(4, -1) (7, 3)$ $m = \frac{4}{3}$ $y+1 = \frac{4}{3}(x-4)$ $3y+3 = 4x-16$ $4x-3y-19 = 0$ as req'd
(b)	2.94	2	
(c)	$c(d-1) - 4(c-d)$ $(c-4)(d-1)$	2	(b) $BC(7,3)(8,1)$ $AD(4,-1)(-1,4)$ $m_1 = -\frac{2}{1} = -2$ $m_2 = \frac{5}{2} = 2.5$ $m_1 \neq m_2$ $BC \nparallel AD$
(d)	$4k+1 = 29$ $k = 7$	2	(c) $DC(1,9) (7,3)$ $m = -\frac{8}{6} = -\frac{4}{3}$ $\text{grad } AC = \frac{4}{3} \rightarrow \text{from (a)}$ $m_1 m_2 = -1$ $DC \perp AC$ $\therefore \angle ACD = 90^\circ$
(e)	$x+3 \leq 1$ or $-x-3 \leq 1$ $x \leq -4$ $-x \leq 4$ $x \geq -4$ 	2	(d) $AC(4,-1) (7,3)$ $d = \frac{\sqrt{(7-4)^2 + (3-(-1))^2}}{\sqrt{25}} = 5$
(f)	$\frac{1}{\sqrt{3}+2} \times \frac{\sqrt{3}-2}{\sqrt{3}-2}$ $= \frac{\sqrt{3}-2}{-1}$ $-\sqrt{3}+2 = a\sqrt{3}+b$ $a = -1$ $b = 2$	2	(e) $d = \frac{ ax_1+by_1+c }{\sqrt{a^2+b^2}}$ $4x-3y-19=0$ $(8,1)$ $d = \frac{ 4 \times 8 - 3 \times 1 - 19 }{\sqrt{16+9}}$ $= \frac{10}{5} = 2$
		(f)	$\text{length } DC (-1,9) (7,3)$ $d = \sqrt{8^2+6^2} = 10$ $\text{Area} = \triangle DCA + \triangle ACB$ $A = \frac{1}{2} \times 10 \times 5 + \frac{1}{2} \times 5 \times 2$ $= 30$

Mathematics Trial, Assessment Task #4, 2002

1 3. (a) i. $7 \times 2(2x-1)^6 = 14(2x-1)^6$.

2 ii. $\frac{5}{4+5x}$.

2 iii. $2x \sin x + x^2 \cos x$.

1 (b) $\int \sec^2 3x dx = \frac{1}{3} \tan 3x + c$.

3 (c) $\int_1^{e^2} \frac{3}{x} dx = 3 [\ln x]_1^{e^2}$
 $= 3 \{2 - 0\}$
 $= 6$.

3 (d) Area $= \int_0^{\frac{1}{2}} e^{2x} dx$
 $= \left[\frac{e^{2x}}{2} \right]_0^{\frac{1}{2}}$
 $= \frac{1}{2} \{e^2 - 1\}$.

2 4. (a) i. $x^2 - 2x + 1 = 8y - 25 + 1$,
 $\therefore (x-1)^2 = 4 \times 2(y-3)$.

1 ii. (1,3).

1 iii. (1,5).

1 iv. $y = 1$.

4 (b) First method: using λ ,
 $x + 2y - 6 + \lambda(x - y) = 0$,
 $(2 - \lambda)y = -(1 + \lambda)x + 6$.
 The slope of line $3x - 2y + 7 = 0$ is $\frac{3}{2}$.
 $\therefore \frac{3}{2} = \frac{\lambda + 1}{\lambda - 2}$,
 $3\lambda - 6 = 2\lambda + 2$,
 $\lambda = 8$.
 $\therefore$ New line is $x + 2y - 6 + 8x - 8y = 0$.
 i.e. $9x - 6y - 6 = 0$,
 $3x - 2y - 2 = 0$.

Second method: first find the intersection,
 $-y = 2y - 6$,
 $y = 2$,
 and $x = 2$.

$\therefore$ Intersection is (2,2).
 The slope of line $3x - 2y + 7 = 0$ is $\frac{3}{2}$.
 $\therefore$ New line is: $y - 2 = \frac{3}{2}(x - 2)$,
 $2y - 4 = 3x - 6$,
 i.e. $3x - 2y - 2 = 0$.

3 (c) Area $\triangle XYZ = \frac{1}{2} \times 4 \times 4$,
 $= 8 \text{ unit}^2$.
 Area of Sector $XYW = \frac{1}{2} \times 4^2 \times \frac{\pi}{4}$,
 $= 2\pi \text{ unit}^2$.
 $\therefore$ Shaded area $= 8 - 2\pi \text{ unit}^2$.

TRIAL 2U HSC 2002 SOLUTIONS

QUESTION 5

a) $y = x^2 - 3x + 6$

(2 marks) $y' = 2x - 3$

at (3,5)

$y' = 2 \times 3 - 3$

$= 3$

hence $m_2 = \frac{-1}{m_1}$
 $= \frac{-1}{3}$

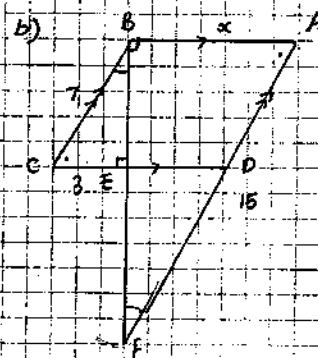
using

$y - y_1 = m(x - x_1)$

$y - 5 = \frac{-1}{3}(x - 3)$

$3y - 15 = -x + 3$

$\therefore x + 3y - 18 = 0$



i) In $\triangle CBE$ and $\triangle AFB$

(3 marks) $\angle CEB = \angle AFB$ (alt. $\angle$ s, $AB \parallel EF$)

$\angle BCE = \angle BAF$ (opp. $\angle$ s, $BC \parallel AF$)

$\angle CBE = \angle AFB$ (sum of $\angle$ s)

$\therefore \triangle CBE \parallel \triangle AFB$ (equiangular)

ii) In $\triangle CBE$ and $\triangle AFB$

(2 marks) $\frac{CB}{CE} = \frac{AF}{AB}$ (corr. sides of $\parallel \Delta$ s)

$\frac{7}{8} = \frac{15}{x}$

$7x = 120$

$x = 17\frac{1}{7}$

$\therefore AB = 17\frac{1}{7} \text{ cm}$

c) $y = x^2 - 1$

(6 marks) $y^2 = (x^2 - 1)^2$
 $= x^4 - 2x^2 + 1$

$V = \pi \int y^2 dx$
 $= 2\pi \int_0^1 (x^4 - 2x^2 + 1) dx$

$= 2\pi \left[\frac{x^5}{5} - \frac{2x^3}{3} + x \right]_0^1$

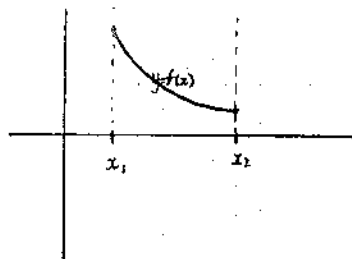
$= 2\pi \left[\left(\frac{1^5}{5} - \frac{2(1)^3}{3} + 1 \right) - \left(\frac{0^5}{5} - \frac{2(0^3)}{3} + 0 \right) \right]$

$= 2\pi \left(\frac{1}{5} - \frac{2}{3} + 1 \right)$

$= 2\pi \left(\frac{8}{15} \right)$

$V = \frac{16\pi}{15} \text{ units}^3$

d) $y = f(x)$ $x_1 \leq x \leq x_2$
 (2 marks) $f(x) > 0, f'(x) < 0, f''(x) > 0$



$f(x)$ above x -axis
 $f'(x)$ negative gradient
 $f''(x)$ concave up

QUESTION 6

a) $y = x^3 + mx + n$ $P(1, 6)$ when $y' = 0$

(3 marks)

$y' = 3x^2 + m$

at $P(1, 5)$

$0 = 3(1)^2 + m$

$\therefore m = -3$

$y = x^3 - 3x + n$

at $P(1, 5)$

$5 = (1)^3 - 3(1) + n$

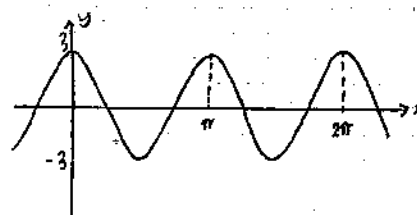
$5 = 1 - 3 + n$

$7 = n$

$\therefore n = 7$

b)

(2 marks)



$y = a \cos mx$

where amplitude $(a) = 3$

$\therefore y = 3 \cos 2x$

period $T = \frac{2\pi}{m}$

$\pi = \frac{2\pi}{m}$

$\therefore m = 2$

c)

(2 marks)

$\ln 2 + 2 \ln 18 - \frac{3}{2} \ln 36$

$= \ln 2 + \ln 324 - \ln 216$

$= \ln \left(\frac{2 \times 324}{216} \right)$

$= \ln 3$

d)

(marks)

$$\sqrt{\frac{1 - \cos^2 \theta}{1 + \tan^2 \theta}}$$

Using $\cos^2 \theta + \sin^2 \theta = 1$

$$\sin^2 \theta = 1 - \cos^2 \theta$$

$$1 + \tan^2 \theta = \sec^2 \theta = \frac{1}{\cos^2 \theta}$$

$$= \sqrt{\frac{\sin^2 \theta}{\frac{1}{\cos^2 \theta}}}$$

$$= \sqrt{\sin^2 \theta \cos^2 \theta}$$

$$= \sin \theta \cos \theta$$

e)

(3 marks)

$$\int_{-1}^7 f(x) dx \approx \frac{h}{3} [(y_0 + y_6) + 4(y_1 + y_5) + 2(y_2 + y_4)]$$

where $h = \frac{b-a}{n}$

$$= \frac{7 - (-1)}{4}$$

$$= 2$$

$$y_0 = 5$$

$$y_1 = 9$$

$$y_2 = 2$$

$$y_3 = -1$$

$$y_4 = -6$$

$$\int_{-1}^7 f(x) dx \approx \frac{2}{3} [(5 - 6) + 4(9 - 1) + 2(2)]$$

$$\approx \frac{2}{3} (-1 + 32 + 4)$$

$$\approx \frac{70}{3}$$

$$\approx 23\frac{1}{3}$$

QUESTION 7

d) i)

$$V = 50000 e^{-0.25t}$$

(1 mark)

when $t = 0$, $V = ?$

$$V = 50000 e^{-0.25 \times 0}$$

$$= 50000$$

$\therefore$ value of new car is \$50000

ii)

when $t = 3$, $V = ?$

(2 marks)

$$V = 50000 e^{-0.25 \times 3}$$

$$= 23618.33$$

$\therefore$ value of car after 3 years is \$23600 (nearest \$100)

iii)

$V = 15000$ $t = ?$

(2 marks)

$$15000 = 50000 e^{-0.25t}$$

$$\frac{15000}{50000} = e^{-0.25t}$$

$$0.3 = e^{-0.25t}$$

$$\ln 0.3 = -0.25t$$

$$t = \frac{\ln 0.3}{-0.25}$$

$$= 4.815$$

$\therefore$ time taken for the car to be worth \$15000 is 4.8 years (nearest tenth of a year).

b) i)
(2 marks)
 $S_n = a + (a+d) + (a+2d)$

where $S_n = 15600$

$a = 1600$

$5600 = 1600 + (1600+d) + (1600+2d)$

$10800 = 3d$

$d = 3600$

$A = \$1600$ $B = \$5200$ $C = \$8800$

ii) $S_n = \frac{a(r^n-1)}{r-1}$
(2 marks)

$S_n = a + ar + ar^2$

$15600 = 1600 + 1600r + 1600r^2$

$14000 = 1600r^2 + 1600r$

$0 = 8r^2 + 8r - 70$

$r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$= \frac{-8 \pm \sqrt{64 + 2240}}{16}$

$\therefore r = \pm 2\frac{1}{2}$

$r = 2.5$

$\therefore A = \$1600$ $B = \$4000$ $C = \$10000$

c) i) $P(2,2) = \frac{2}{7} \times \frac{1}{6}$
(1 mark)
 $= \frac{1}{21}$

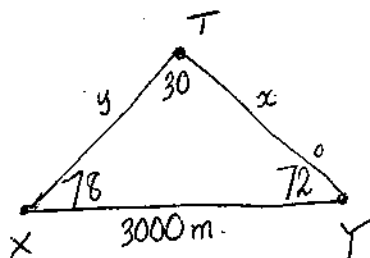
ii) $P(1+6) = \frac{1}{7} \times \frac{4}{6}$
(2 marks)
 $= \frac{4}{42}$

$P(6+6) = \frac{4}{7} \times \frac{3}{6}$
 $= \frac{12}{42}$

$P(6+4) = \frac{4}{7} \times \frac{1}{6}$
 $= \frac{4}{42}$

$P(1+8) = \frac{8}{42} + \frac{12}{42}$
 $= \frac{20}{21}$

(a)



$$\frac{y}{\sin 72^\circ} = \frac{3000}{\sin 30^\circ}$$

$$y = \frac{3000 \times \sin 72^\circ}{\sin 30^\circ}$$

$$\therefore 5706 \text{ m.} \div 5.706 \text{ km.}$$

(3)

$$(b) mx^2 - 20x + m = 0$$

$$\text{let } x = -\frac{1}{2} \quad \frac{m}{4} + 10 + m = 0$$

$$m + 40 + 4m = 0$$

$$5m = -40$$

$$m = -8$$

$$\text{So we have } -8x^2 - 20x - 8 = 0$$

$$\text{and } -\frac{1}{2} + \alpha = -\frac{20}{-8}$$

$$r(2x+1)(x+2) = 0$$

$$x = -\frac{1}{2}, x = -2$$

$$\alpha = -\frac{20}{8} + \frac{1}{2}$$

$$= -2$$

Other root is -2.

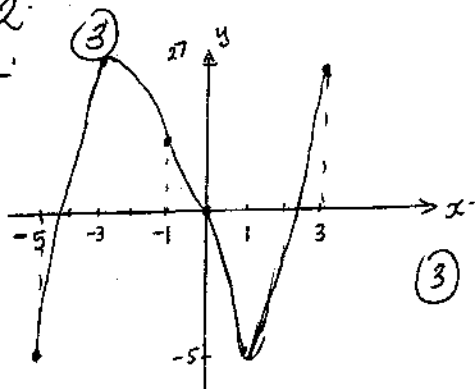
$$(c) y = x^3 + 3x^2 - 9x$$

$$y' = x(x^2 + 3x - 9)$$

$$y' = 3x^2 + 6x - 9$$

$$= 3(x^2 + 2x - 3)$$

$$y'' = 6x + 6 = 6(x+1)$$



$$3(x^2 + 2x - 3) = 0$$

$$x^2 + 2x - 3 = 0$$

$$(x+3)(x-1) = 0$$

$$x = -3, x = 1$$

$$y = 27, y = -5$$

stat points are
(-3, 27) and (1, -5)

(2)

(ii) Curve is concave
upwards when $y'' > 0$

$$6(x+1) > 0$$

$$x+1 > 0$$

$$x > -1 \quad (1)$$

$$(iii) (-3, 27) y'' < 0 \text{ max}$$

$$(1, -5) y'' > 0 \text{ min}$$

(-1, 11) inflection

$$\text{At } x = -5, y = -5$$

$$x = 3, y = 27$$

$$(a) |x+1| = 2x+7$$

$$x+1 = 2x+7$$

$$-6 = x$$

$$\text{LHS} \neq \text{RHS}$$

$$x+1 = -2x-7$$

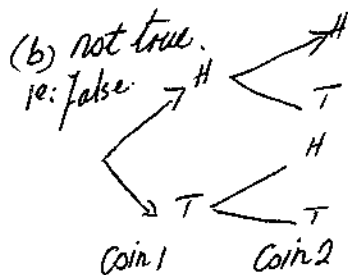
$$3x = -8$$

$$x = -\frac{8}{3}$$

$$\text{LHS} = 1\frac{2}{3}$$

$$\text{RHS} = 1\frac{2}{3}$$

$$x = -\frac{8}{3} = -2\frac{2}{3} \quad (-2.6) \quad (3)$$

4 possibilities
not 3. (2)

$$(c) (i) a = 2\pi^2 \cos \pi t$$

$$\text{data } t=0$$

$$x=0$$

$$V=\pi$$

$$V = \int 2\pi^2 \cos \pi t \, dt$$

$$= \frac{2\pi^2}{\pi} \int \pi \cos \pi t \, dt$$

$$V = 2\pi \sin \pi t + C_1$$

$$\pi = C_1$$

$$V = 2\pi \sin \pi t + \pi$$

$$x = \int (2\pi \sin \pi t + \pi) \, dt$$

$$x = -2 \cos \pi t + \pi t + C_2$$

(ii) when $V=0$

$$2\pi \sin \pi t + \pi = 0$$

$$2\pi \sin \pi t = -\pi$$

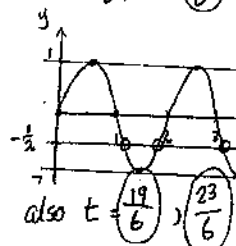
$$\sin \pi t = -\frac{\pi}{2\pi} = -\frac{1}{2}$$

$$\sin \pi t = -\frac{1}{2}$$

$$\pi t = \frac{7\pi}{6}, \frac{11\pi}{6}$$

$$t = \frac{7\pi}{6\pi} = \frac{7}{6} \text{ sec}$$

$$t = \frac{11\pi}{6\pi} = \frac{11}{6}$$



$$(iii) t = \frac{19}{6} \quad x = -2 \cos \frac{19\pi}{6}$$

$$= -2 \times \frac{\sqrt{3}}{2} + \frac{19\pi}{6}$$

$$= 2 + \sqrt{3} + \frac{19\pi}{6}$$

$$t = \frac{23}{6} \quad x = -2 \cos \frac{23\pi}{6} + \frac{23\pi}{6}$$

$$= -2 \times \frac{\sqrt{3}}{2} + \frac{23\pi}{6}$$

$$= 2 + \sqrt{3} + \frac{23\pi}{6}$$

$$\text{At } t = \frac{19}{6} \quad x \text{ is large}$$

$$\text{so } x = -2 \cos \pi t + \pi t$$

$$② C(n) = 1000(0.8)^n + 10,000[1 + .8 + .8^2 + \dots + .8^n]$$

$$\text{as } n \rightarrow \infty \quad 1000(0.8)^n \rightarrow 0$$

$$\text{now } 10,000[1 + .8 + .8^2 + \dots + .8^{n-1}]$$

$$\left. \begin{array}{l} \text{GP } a=1 \\ r=.8 \\ n=n \end{array} \right\} S_n = \frac{1(.8^n - 1)}{.8 - 1} = \frac{.8^n - 1}{-.2} = \frac{1 - .8^n}{.2}$$

$$\text{so we have } 10,000 \times 5(1 - .8^n) = 50,000(1 - .8^n) \quad (3)$$

$$\text{as } n \rightarrow \infty \quad 50,000(1 - .8^n) \rightarrow 50,000$$

$$⑥ (i) \phi(t) = A(1+t)e^{-0.5t}$$

$$\phi(t) = Ae^{-0.5t} + Ate^{-0.5t}$$

$$\begin{aligned} \phi'(t) &= -0.5Ae^{-0.5t} + At \times -0.5e^{-0.5t} + Ae^{-0.5t} \\ &= 0.5Ae^{-0.5t} - 0.5Ate^{-0.5t} \end{aligned}$$

$$\begin{aligned} \phi''(t) &= -0.25Ae^{-0.5t} = (0.5At \times -0.5e^{-0.5t} + 0.5Ae^{-0.5t}) \\ &= -0.25Ae^{-0.5t} + 0.25Ate^{-0.5t} - 0.5Ae^{-0.5t} \\ &= -0.75Ae^{-0.5t} + 0.25Ate^{-0.5t} \end{aligned}$$

$$\begin{aligned} &4(-0.75Ae^{-0.5t} + 0.25Ate^{-0.5t}) + 4(0.5Ae^{-0.5t} - 0.5Ate^{-0.5t}) + Ae^{-0.5t} + Ate^{-0.5t} \\ &= 0 \quad (4) \end{aligned}$$

$$v) (ii) \psi(0) = 10$$

$$\text{So when } t=0 \quad A(1+t)e^{-0.5t} = 10$$

$$A(1+0)e^{-0.5 \times 0} = 10$$

$$A = 10$$

$$\text{So } \phi(t) = 10(1+t)e^{-0.5t}$$

$$\begin{aligned} \phi'(t) &= 0.5 \times 10e^{-0.5t} - 0.5 \times 10te^{-0.5t} \\ &= 5e^{-0.5t} - 5te^{-0.5t} \end{aligned}$$

$$= 5e^{-0.5t}(1-t)$$

$$\text{let } \phi'(t) = 0 \Rightarrow t = 1$$

$$\begin{aligned} \text{Sub } t=1, \quad \phi''(t) &= -0.75 \times 10e^{-0.5} + 0.25 \times 10 \times 1e^{-0.5} \\ &= -7.5e^{-0.5} + 2.5e^{-0.5} \\ &= -5e^{-0.5} < 0 \end{aligned}$$

at $t=1$, $\phi(t)$ is a max.

$$\begin{aligned} \text{When } t=1, \quad \phi(t) &= 10(2)e^{-0.5} \\ &= \frac{20}{e^{0.5}} \approx 12.13 \quad (4) \end{aligned}$$

$$\begin{aligned} (iii) \phi(t) &= 10(1+t)e^{-0.5t} \\ &= 10e^{-0.5t} + 10te^{-0.5t} \\ &= \frac{10}{e^{0.5t}} + \frac{10t}{e^{0.5t}} \quad \text{as } t \rightarrow \infty \quad \phi(t) \rightarrow 0 \quad (i) \end{aligned}$$